MADAN MOHAN MALAVIYA UNIVERSITY OF TECHNOLOGY

B.TECH. COMPUTER SCIENCE & ENGINEERING / ALLIED BRANCHES

END SEMESTER EXAMINATION (2024-25) -- SEMESTER 7

COURSE: CRYPTOGRAPHY & INFORMATION SECURITY (BCS-702)

Time Allowed: 3 Hours Maximum Marks: 70 Marks Paper Code: BCS-702

Roll Number: [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] | Total Printed Pages: 2

NOTE: Attempt all Sections. Assume suitable missing data if any. Programmable calculators are NOT permitted.

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SECTION A: Short Answer Type Questions (Attempt ALL questions) [10 x 2 = 20 Marks]

Q1. (a) State the CIA triad in information security.

Q1. (b) Differentiate between active attacks and passive attacks with examples.

Q1. (c) What is difference between stream cipher and block cipher?

Q1. (d) Explain avalanche effect in symmetric cryptographic block ciphers.

Q1. (e) What is Feistel cipher structure? Name ciphers based on Feistel network.

Q1. (f) What is Fermat Little Theorem and Euler Totient Function?

Q1. (g) What is discrete logarithm problem in Diffie-Hellman key exchange?

Q1. (h) Define cryptographic hash function properties: Pre-image, Second pre-image, Collision resistance.

Q1. (i) What is difference between Message Authentication Code (MAC) and Digital Signature?

Q1. (j) Explain role of Certificate Authority (CA) in Public Key Infrastructure (PKI).

SECTION B: Analytical & Computational Problems (Attempt any THREE) [3 x 10 = 30 Marks]

Q2. (a) Explain Data Encryption Standard (DES) architecture: 16 Feistel rounds, S-Box substitution, and key schedule.

Q2. (b) Explain Advanced Encryption Standard (AES) round operations: SubBytes, ShiftRows, MixColumns, AddRoundKey.

Q2. (c) Perform RSA encryption and decryption for primes p = 11, q = 13, encryption key e = 7, and message M = 9.

Q2. (d) Explain Diffie-Hellman Key Exchange algorithm with prime q = 23, primitive root alpha = 9, private keys X_A = 4, X_B = 3.

Q2. (e) Explain SHA-512 cryptographic hash function architecture, compression function, and message padding.

SECTION C: Long Technical & Design Questions (Attempt any ONE from each) [5 x 10 = 50 Marks]

Q3. (a) Explain Elliptic Curve Cryptography (ECC), point addition, point doubling, and discrete logarithm. [10 Marks]

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(b) Explain Digital Signature Standard (DSS / DSA) signing and verification equations. [10 Marks]

Q4. (a) Explain Block Cipher Modes of Operation: ECB, CBC, CFB, OFB, CTR with security analysis. [10 Marks]

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(b) Explain Man-in-the-Middle attack on Diffie-Hellman and mitigation using digital certificates. [10 Marks]

Q5. (a) Explain SSL / TLS handshake protocol step-by-step with cipher negotiation and session keys. [10 Marks]

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(b) Explain IPsec architecture: Authentication Header (AH) vs Encapsulating Security Payload (ESP) in Transport and Tunnel mode

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Q6. (a) Explain Kerberos authentication service protocol with Authentication Server (AS) and Ticket Granting Server (TG

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(b) Explain Pretty Good Privacy (PGP) for email security: Confidentiality, integrity, and authentication. [10 Marks]

Q7. (a) Explain Firewall types: Packet filtering, Stateful inspection, Application gateway, and DMZ design. [10 Marks]

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(b) Explain Intrusion Detection Systems (IDS): Signature-based vs Anomaly-based detection. [10 Marks]

EXAMINATION EVALUATION SCHEME & RUBRIC GUIDELINES:

1. Section A: Direct definition, time complexity analysis, formula derivation, or syntax verification.
2. Section B: Step-by-step mathematical proof, trace tables, algorithmic pseudocode, and state diagrams.
3. Section C: Architectural design, full code implementation, edge case handling, and space/time tradeoffs.

***** END OF QUESTION PAPER -- DEVENGERS ACADEMIC VAULT *****